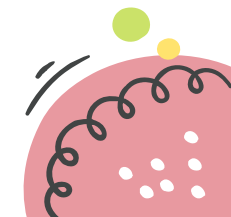




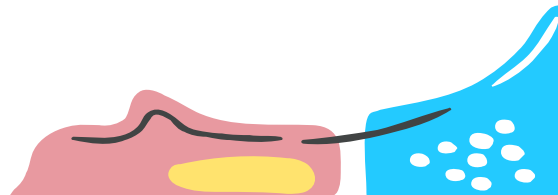
Overview of Learning, Education, and Families

By: Sarah Oide and Victoria Hays



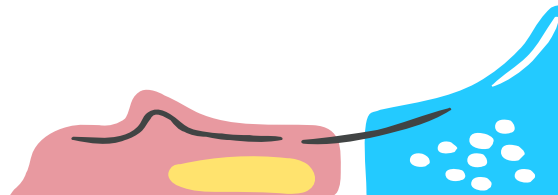


Today's Class

- Guiding Questions
 - Contributing Disciplines
 - History at CHI
 - Scholars
 - Framing Papers
 - Questions
- 
- 



Guiding Questions

- How to design, deploy, and evaluate technologies for **diverse learning contexts** and educational settings?
 - How do **children, parents, and families** interact with existing and emerging technologies?
 - How might technologies support **equitable and inclusive** learning experiences?
- 
- 



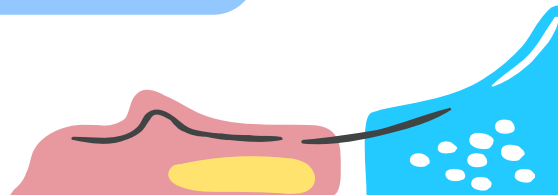
Guiding Questions Cont.

Where?

Formal (school) and informal settings (museums, after school, home, game-based, online)

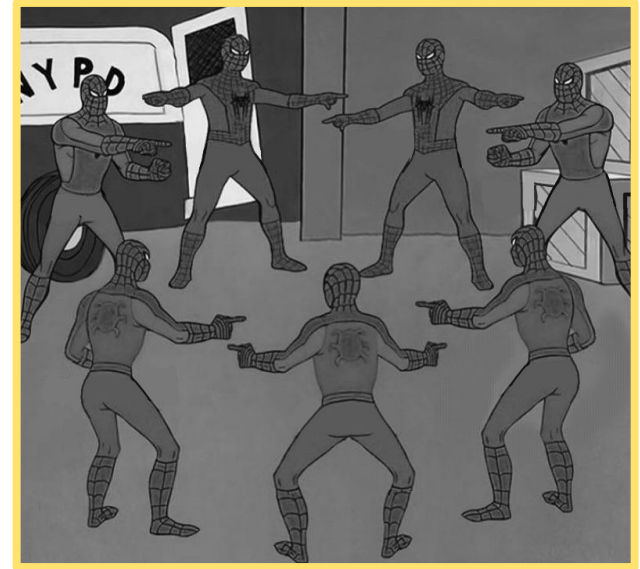
Who?

Learners across the lifespan (children, parents, families, educators, students, hobbyists, etc.)



Contributing Disciplines

- Education
- Learning Sciences
- Psychology
- Art and Design
- Media and Communication Studies
- Sociology
- Data Science





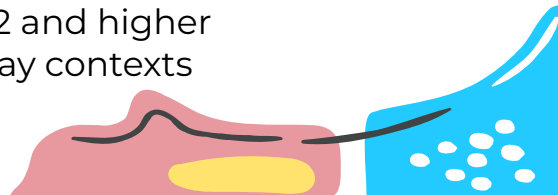
History of CHI and Future Directions

(Pammer-Schindler et al., 2020)

Rapid Growth in late 2010s

- Learning technologies
- Educational equity research
- Child-Computer Interaction (CCI)
- Shifts from a special application topic → workshop → subcommittee

CHI LEF Inaugural Workshop → Subcommittee

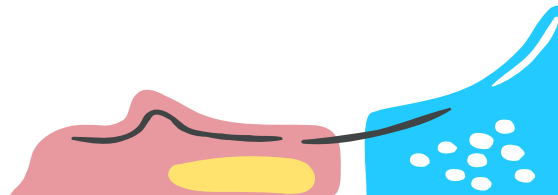
- 179 submissions in the first year, organized around three guiding questions, “What does learning mean to us? What are valid ways of evaluating a learning contribution? How can we foster the relationship between learning and HCI?”
 - In 2024 234 submissions and In 2025 345 submissions
 - This reflects the growth of educational technologies in k-12 and higher ed, rise in AI and learning analytics, and learning in everyday contexts
- 
- 



Relevant Venues

Interaction Design and Children(IDC): conferences focused on researching, designing, and evaluating new technologies and tools for children.

International Society of the Learning Sciences (ISLS): interdisciplinary society dedicated to the research of learning in all forms.





IDC Framing

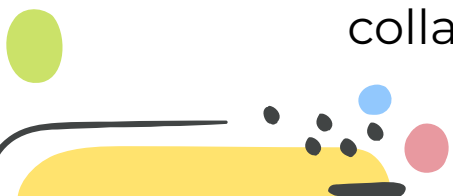
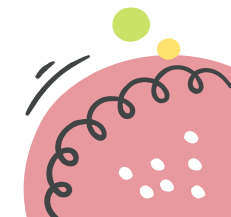
(Kawas et al., 2020)



Core Areas of Interest

- Designing for sustainability, equity, diversity and inclusion
- Innovative interactive technologies for children 0-18
- Designing for and with children
- Increasing focus on AI policy (clear justification for AI use, demonstrated benefits for children, meaningful contributions to the IDC community)

Evolving Research Trends

- Growing emphasis on designing and evaluating in authentic contexts
 - Increased value on empirical research
 - Rise in themes of pro social interaction and positive social collaboration
- 
- 

Scholars



Allison Druin



Saba Kawas

Cooperative Inquiry

- A theoretical framework for research and design
- Advocates for intergenerational design teams to develop new technologies for children, with children





Cooperative Inquiry: Commitments

Full Participation

Children as full partners in research and design

Context Matters

Knowledge is situated in the field (real-world contexts)

Prototyping

Visualizing ideas through low-tech and high-tech designs





Cooperative Inquiry: Methods

Contextual Inquiry

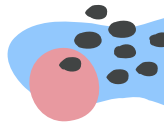
Adapted for children during field research and analysis; children and adults treated as researchers

Participatory Design

Prototyping collaboratively while remaining mindful of socialized dynamics

Technology Immersion

Study for broader periods with high volume of tech



Cooperative Inquiry: Methods

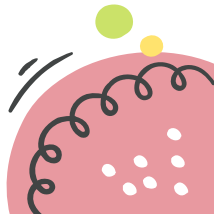
RAW DATA:			DATA ANALYSIS:		
Time	Quotes	Activities	Activity Patterns	Roles	Design Ideas
39:20	"I want the playing one."	Child clicks on the scared cat and tries to take out another one. It doesn't work.	Difficulty with mouse dragging.		Look for alternative input devices or don't use dragging with a mouse.
39:50	"Awww. The kitten was afraid."	Child clicks on another basket with a cat.	Tells stories about actions on screen.	Storyteller	Offer children storytelling opportunities with technology.
40:20	"Which one's the playful one?"	Child looks for a playful cat.	Child knows what she likes.	Searcher	
41:00	"I don't want to name my kitty."	Child doesn't name her cat when prompted to by the computer.	Child knows what she likes.		
41:30	"That's to give milk."	Child clicks on different icons to see what they do.	Tests out what can be done.	Explorer	Make technology easy to explore.

Table 1: Portion of a contextual inquiry diagram created by adults



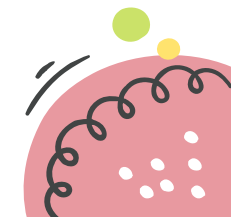
Cooperative Inquiry: Takeaways

- Considering **power dynamics** across the research process
- Beyond technology as the outcome of a design process; research as fostering learning through a **community of practice**
- Remaining flexible and humble to methodological challenges





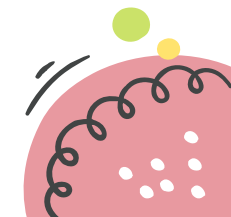
Questions from Perusall

1. What happened to CHIKids?
 2. Ethical considerations when involving children in research, past and present?
 3. Is cooperative inquiry different from participatory design?
- 
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Discussion of Learning, Education, and Families

By: Sarah Oide and Victoria Hays



Scholars



Betsy DiSalvo



Tiffany Do



Valentina Andries

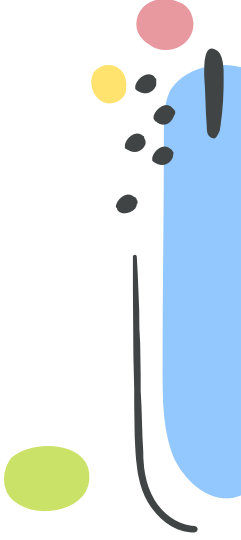
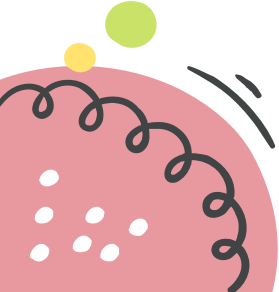


Viktoria Pammer-Schindler



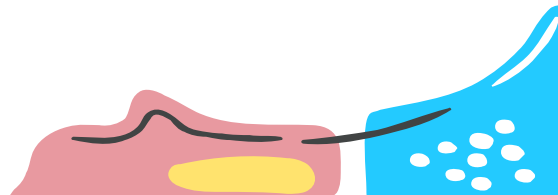
The Secret Lives of Children's Money

Valentina Andries, Cara Wilson, Harvey Everson,
Caroline McKinley, Chris Elsdon





Review: Guiding Questions

- How to **design, deploy, and** evaluate technologies for **diverse learning contexts** and educational settings?
 - How do **children, parents, and families** interact with existing and emerging technologies?
 - How might technologies support **equitable and inclusive** learning experiences?
- 
- 

Why is This Worthy of Study?

- Adults' perspectives prioritized in financial literacy literature
 - Children's conceptualizations overlooked
- Youth navigate an increasingly digital financial landscape





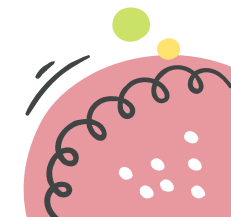
Areas of Inquiry

Within HCI

Child-Computer Interaction;
Financial literacy technologies

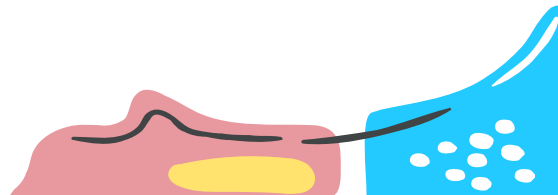
Beyond HCI

Sociology (sociocultural meaning
of money - power, capital, morality)

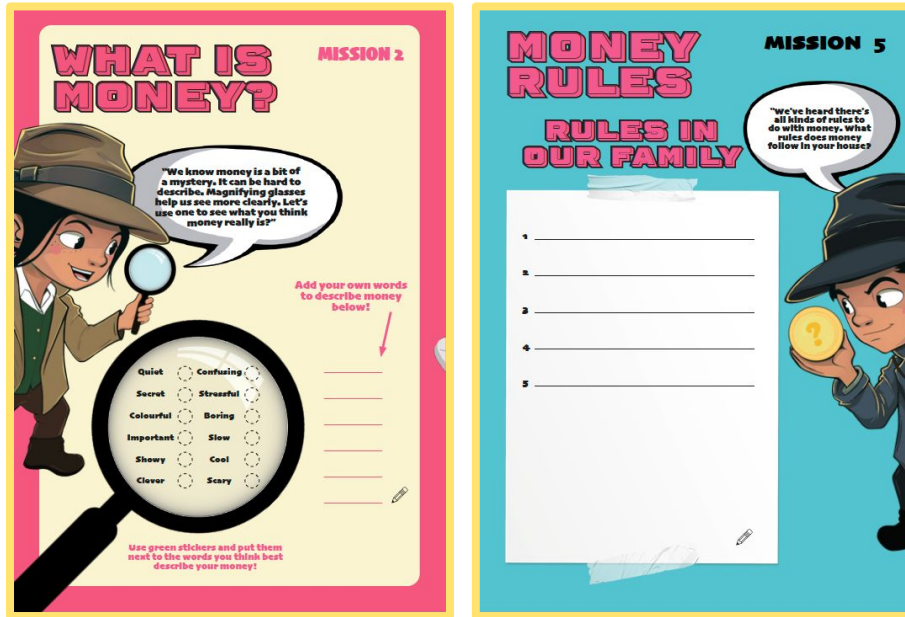




Findings

- Children's monetary journey; cash to digital economy as a major **milestone** in development of agency
 - Children encounter **non-real**, digital currencies, suggesting multiple sites for financial education
- 
- 

Research Contribution



- Empirical
 - Cultural probes with accompanying artifact data and interview analysis



Figure 1: 'Money Mysteries' booklet and other items contained in the 'Detective Probe Pack'



Takeaways

Design

Differentiating between types of digital money and its finite nature

Currency Tensions

Managing the transition between physical and digital currencies can be challenging for parents

Sociological

Understandings of money are shaped by social norms, expectations, and relationships

Games

An important context where children are making sense of digital currencies



Takeaways Cont.

 Cymraeg

BenefitsEveryday moneyFamily & careHomesMoney troublesPensions & retirementSavingsWork

Home > Family & care > Talk money > Talk Learn Do

Learning about money by age

Talk Learn Do

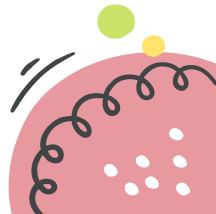
- How to talk to your children about money >
- Learning about money by age
- Shopping trips with children >
- How to handle pestering >
- Pocket money and saving >
- Explaining digital money to children >

You can start to teach children about money from an early age. Children as young as three are ready to learn the basics. Here is a guide to what children are likely to understand, based on their age.



What's in this guide

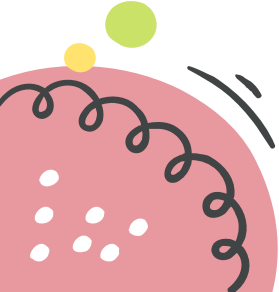
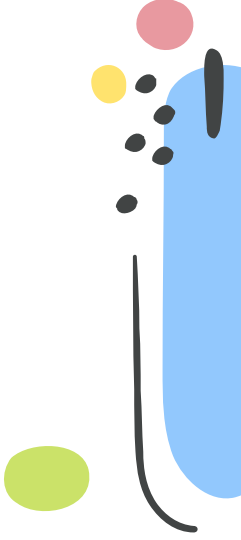
- Three to four-year-olds ↓
- Five to six-year-olds ↓
- Seven to eight-year-olds ↓

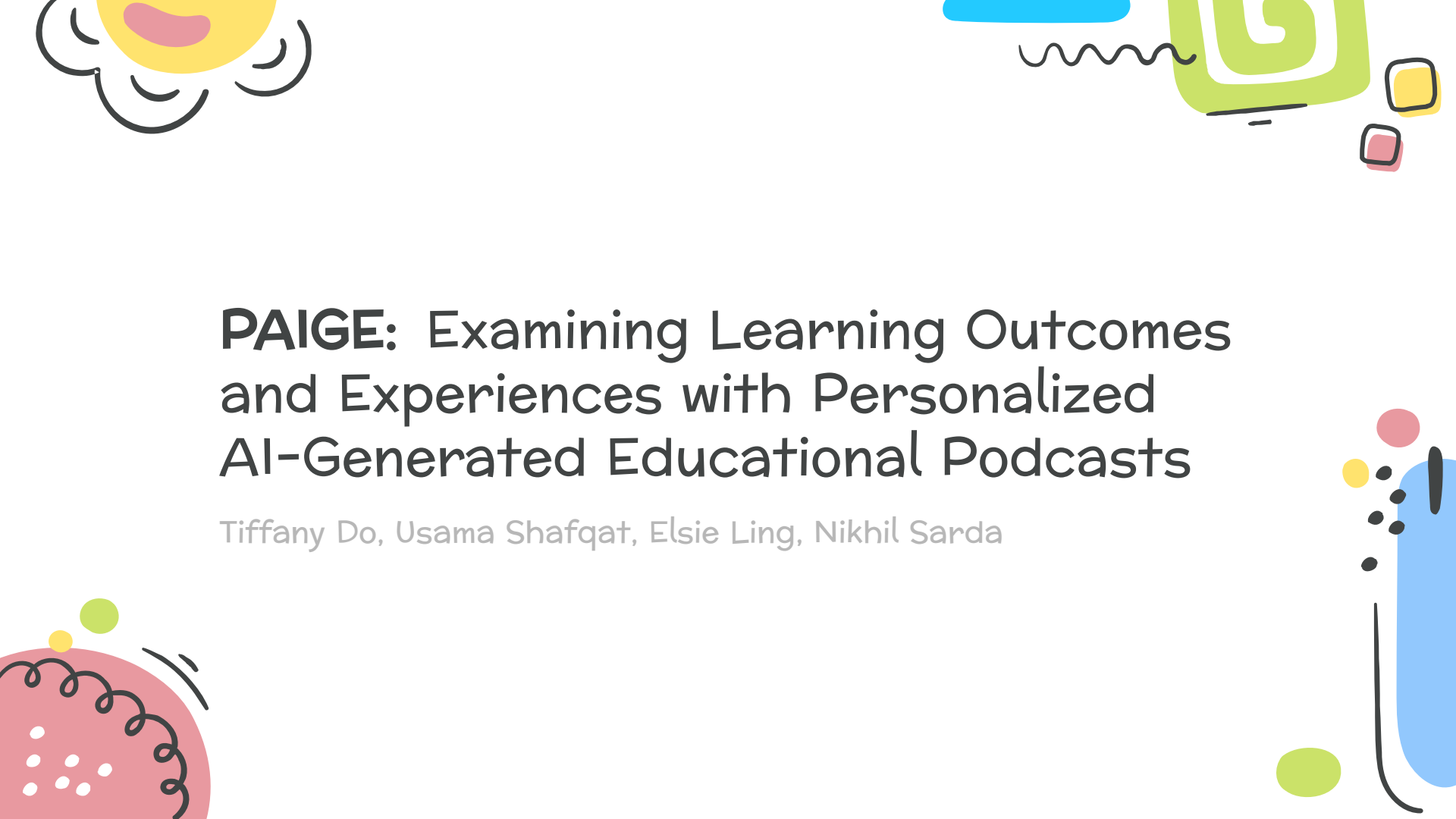




What reactions did you have to this paper and its findings?

Why was this a “Best Paper”?

- The topic is easily sold as important, unique, and understudied in the field (e.g., a universal issue for both adults and children) **AND** the method is awesome - the design itself is persuasive
 - Was this cultural probes method necessary to answer the RQs? Maybe, but there are other aspects of knowledge that may be captured by the project through play-based approach
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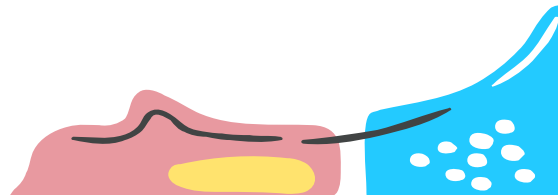


PAIGE: Examining Learning Outcomes and Experiences with Personalized AI-Generated Educational Podcasts

Tiffany Do, Usama Shafqat, Elsie Ling, Nikhil Sarda



Review: Guiding Questions

- How to design, deploy, and evaluate technologies for **diverse learning contexts** and educational settings?
 - How do **children, parents, and families** interact with existing and emerging technologies?
 - How might technologies support **equitable and inclusive** learning experiences?
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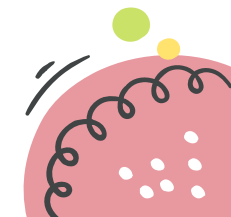
Areas of Inquiry

Within HCI

AI-Generated Educational Content
(e.g., storytelling, video)

Beyond HCI

Education (Personalized Learning
Contexts and Content)



Why is This Worthy of Study?

- Podcasts as a preferred study method
- “Traditional textbooks are becoming increasingly challenging for students to engage with” (p. 1)

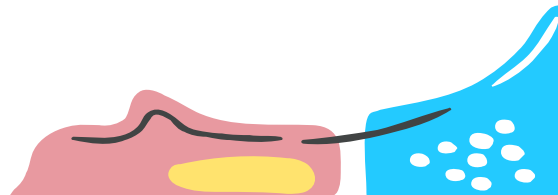


Table 1: Means (M) and standard deviations (SD) of all dependent variables, disaggregated by subject and content style.

	Content Type	Test Score M (SD)	Attractiv. M (SD)	Stimul. M (SD)
Phil.	Generalized	5.85 (1.35)	5.26 (0.67)	4.86 (1.05)
	Personalized	7.30 (1.45)	5.15 (0.84)	4.95 (1.05)
	Textbook	5.95 (1.57)	4.58 (1.23)	4.42 (1.31)
Psych.	Generalized	4.75 (1.62)	5.29 (1.17)	4.96 (1.34)
	Personalized	5.70 (1.38)	5.63 (1.06)	5.32 (1.24)
	Textbook	4.80 (1.57)	4.30 (1.05)	4.45 (1.04)
Gov.	Generalized	6.10 (1.45)	5.22 (0.75)	4.63 (0.81)
	Personalized	6.70 (1.38)	4.98 (1.19)	4.64 (1.25)
	Textbook	7.15 (1.27)	4.67 (0.68)	4.68 (0.89)



Findings

- Podcasts **more** enjoyable than textbooks (regardless of personalization)
 - **Personalized** podcasts had greater learning outcomes than general podcast or textbook
 - Subjects matter: No interaction for government
- 
- 



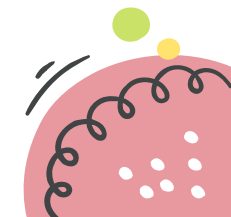
Research Contribution

Empirical

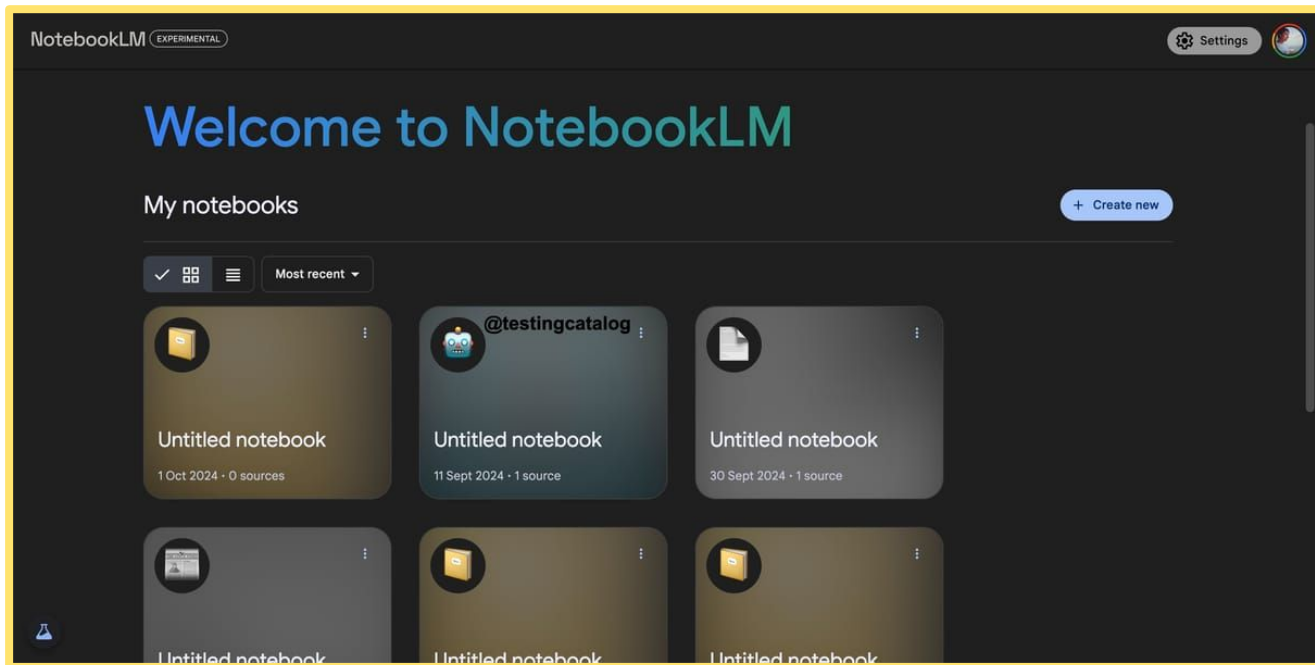
Testing learning experiences
with varied AI-generated
educational content

Artifact

Novel system/tool to produce
AI-generated podcasts



A Contribution to Industry?





Takeaways

Personalized Learning

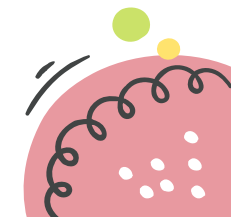
Efficacy of personalized, AI-driven learning experiences

Subject Matter

Differences in efficacy of digital learning tools may vary based on subject

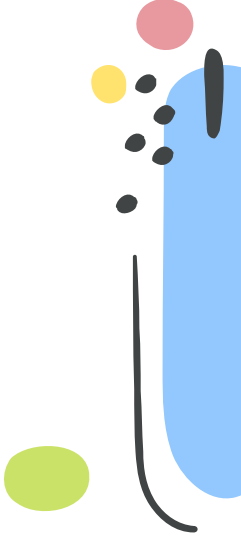
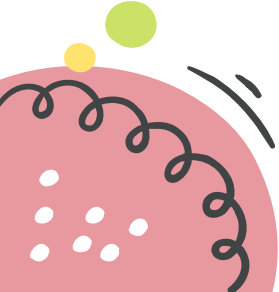
Design

Integrating visual elements into multimodal, AI-driven learning experiences



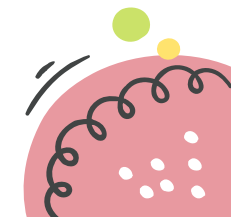


What feedback did you have
about this paper?





Class Feedback from Perusall

- We've heard this message about EdTech before...
 - Instrument for “knowledge measurement”
 - Scope of subject matter (no math, Physics) and implementation
 - Limited discussion of students who *disliked* the AI-generated content
- 
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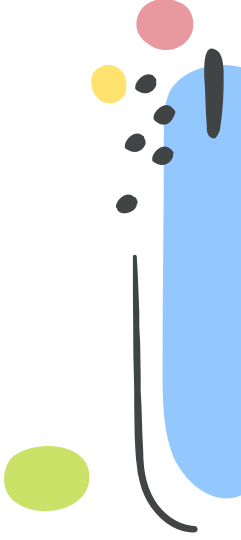
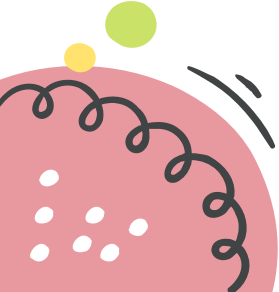
Broader Context: Critical Perspectives

- Contested nature of EdTech
 - Labor and maintenance of deployed technologies
- Technology as a bandaid for structural, systemic challenges in education and pedagogy





Why was this a “Best Paper”?





Cross-Paper Discussion

- What, if anything, stands out to you about these papers versus other subfields?
 - Each subfield emphasizes different research contributions/areas
 - Questions?
- 
- 